

NMJSim - Neuromuscular Junction Simulation

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NMJSim V2.0

Neuromuscular Junction Membrane Potential Simulation

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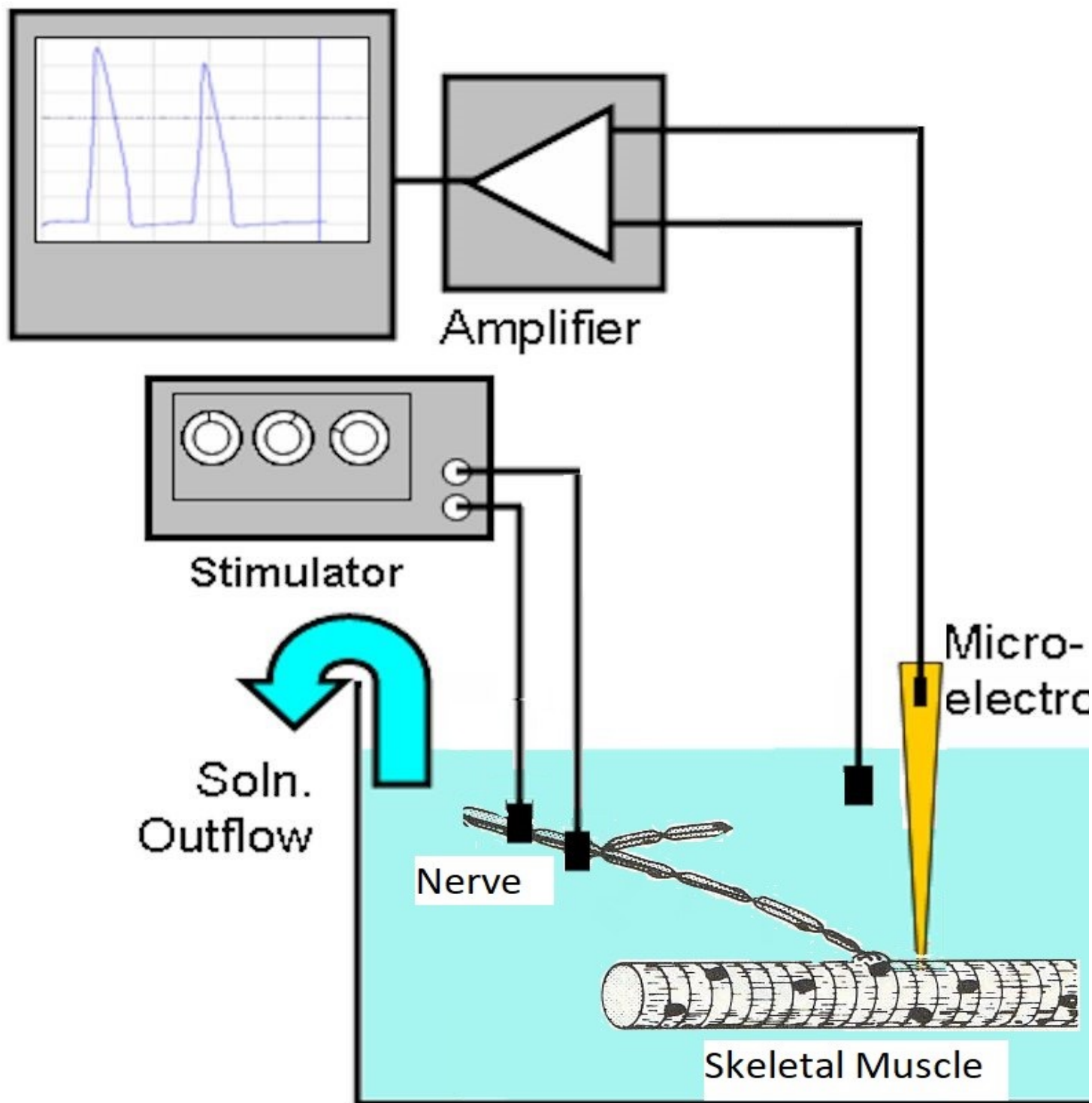
Introduction >

NMJSim is a simulated experiment for investigating the effects of a variety of drugs and ions on neuromuscular transmission at the mammalian neuromuscular junction.

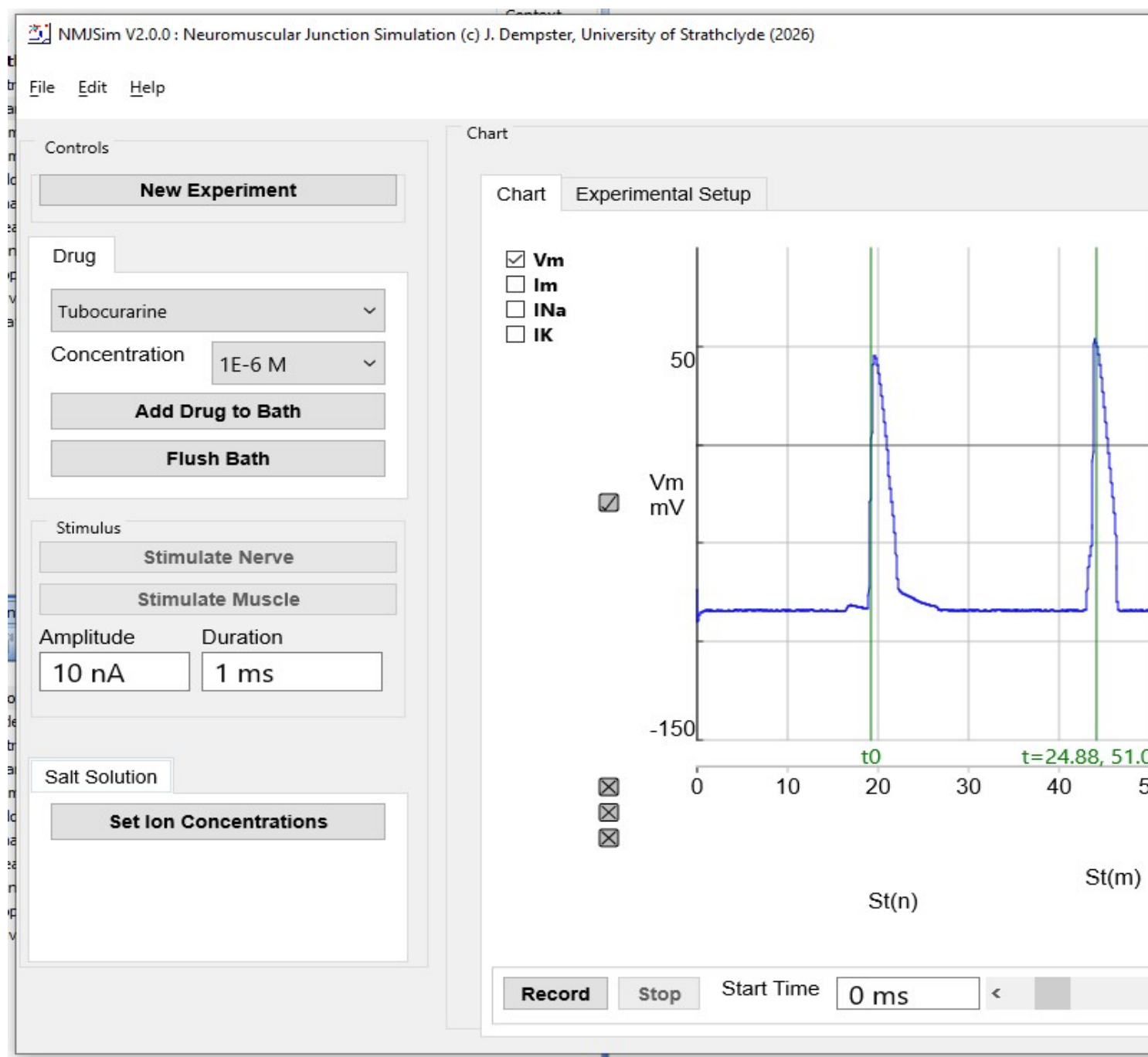
The intracellular membrane potential of a skeletal muscle cell is recorded using an amplifier via a glass micropipette electrode inserted into the cell close to the neuromuscular junction and connected to an oscilloscope recording device. The muscle cell can also be stimulated by injecting a small current into the cell via the same electrode. The nerve supplying the neuromuscular junction can be also be stimulated by applying a small voltage across a pair of electrodes upon which the nerve is lying.

Drugs can be applied to the bath and also concentration of the Na, Ca, Mg and K ions in the bathing medium changed.

Neuromuscular Junction Action Potential Recording Setup



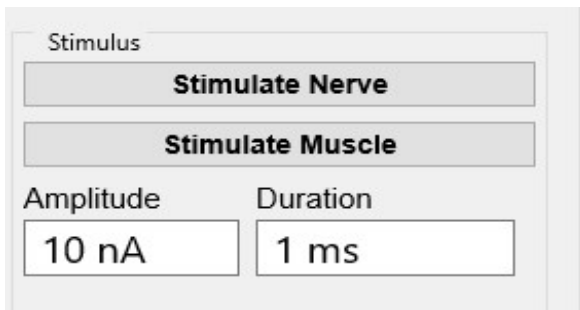
Starting an Experiment >



To start a new experiment:

- Click the **New Experiment** button.
- Click the **Record** button to start the chart recorder running.
- Select and apply a drug from the **Drugs** list OR change the salt solution in the bath.
- Click the **Stimulate Nerve** button to stimulate the nerve OR **Stimulate Muscle** to stimulate the muscle cell directly.

Stimulating the Nerve or Muscle >



The image shows a software interface for stimulus control. It has a title bar 'Stimulus'. Below it are two buttons: 'Stimulate Nerve' and 'Stimulate Muscle'. Below these buttons are two input fields: 'Amplitude' with the value '10 nA' and 'Duration' with the value '1 ms'.

Nerve

To stimulate the nerve leading to the neuromuscular junction:

- a) Click the **Stimulate Nerve** button to apply the stimulus.

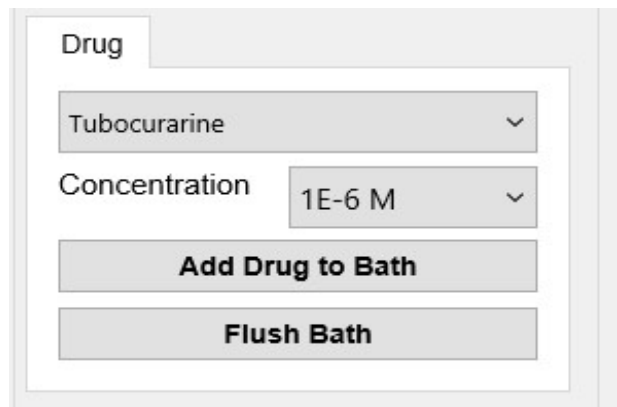
Muscle

To inject a current into the muscle cell via the microelectrode:

- a) Enter the amplitude (nA) of the stimulus current into the stimulus **Amplitude** box.
- b) Enter the duration (ms) of the stimulus current pulse into the stimulus **Duration** box.
- c) Click the **Stimulate Muscle** button to apply the stimulus.

Adding Drugs >

To add a drug to the bath:



The screenshot shows a software interface for adding drugs to a bath. It features a tab labeled 'Drug'. Below the tab, there is a dropdown menu currently displaying 'Tubocurarine'. To the right of this menu is a 'Concentration' label and another dropdown menu showing '1E-6 M'. Below these elements are two rectangular buttons: 'Add Drug to Bath' and 'Flush Bath'.

- Select the type of drug to be applied from the **Drug** list.
- Select the concentration (in M/litre) of the drug to be added from the **Concentration** box. (Note. The notation "E" stands for "x 10⁻⁶", i.e. 1E-6 M indicates 1 x 10⁻⁶ M)
- Click the **Add Drug to Bath** button to add the drug to the bath.
- After observing the effects of a drug you can remove it (and all other drugs that might have previously been added) from the bath by clicking **Flush Bath**.

Drug Properties

d-Tubocurarine: A competitive antagonist at nicotinic cholinceptors.

Neostigmine: An antagonist of the cholinesterase enzymes which break down acetylcholine in the synaptic cleft.

Suxamethonium: A nicotinic cholinceptor agonist (also known as succinylcholine).

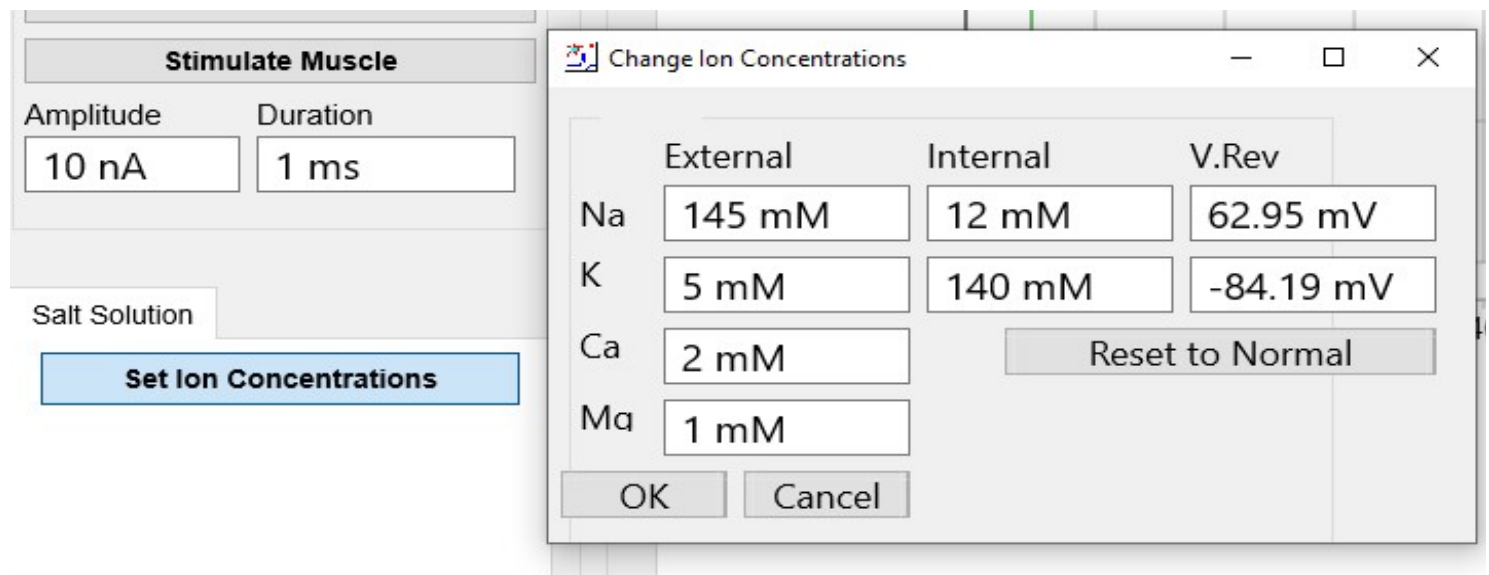
Tetrodotoxin: A potent neurotoxin, derived from the pufferfish, which blocks voltage-activated sodium channels.

3,4-Diaminopyridine: A blocker of voltage-activated potassium channels.

μ -Conotoxin: One of a group of neurotoxic peptides isolated from the venom of the marine cone snail, *Conus*. It is a selective blocker of muscle specific, voltage-activated sodium channels.

Changing the Salt Solution >

To change the concentration of sodium or potassium ions in the bath:



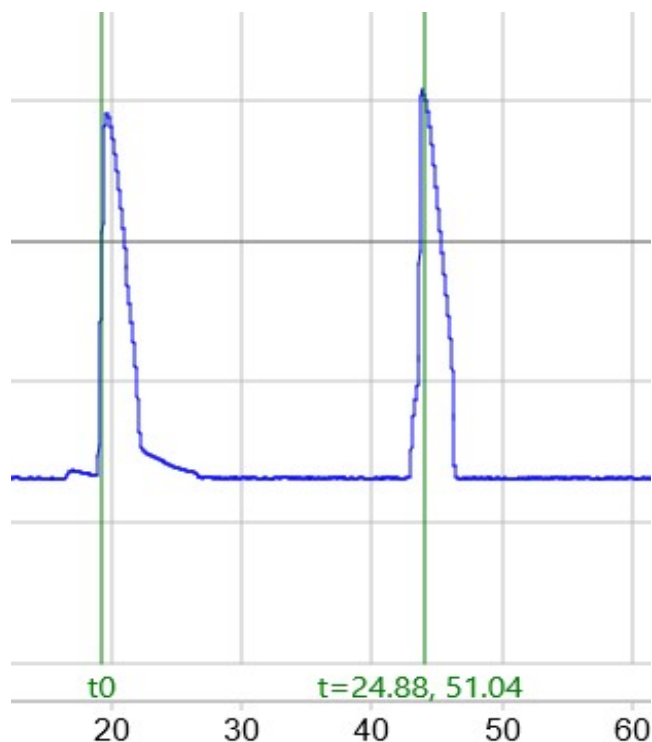
- Click the **Set Ion Concentrations** button.
- Enter the new ion concentration, Sodium (Na), Potassium (K), Calcium (Ca), or Magnesium (Mg), in mM/l in the appropriate box. (Note. Only External ion concentrations can be changed.)
- Click the **OK** button to change the salt solution in the bath to the new solution.

Measuring the Membrane Potential >

To measure the recorded skeletal muscle membrane potential:



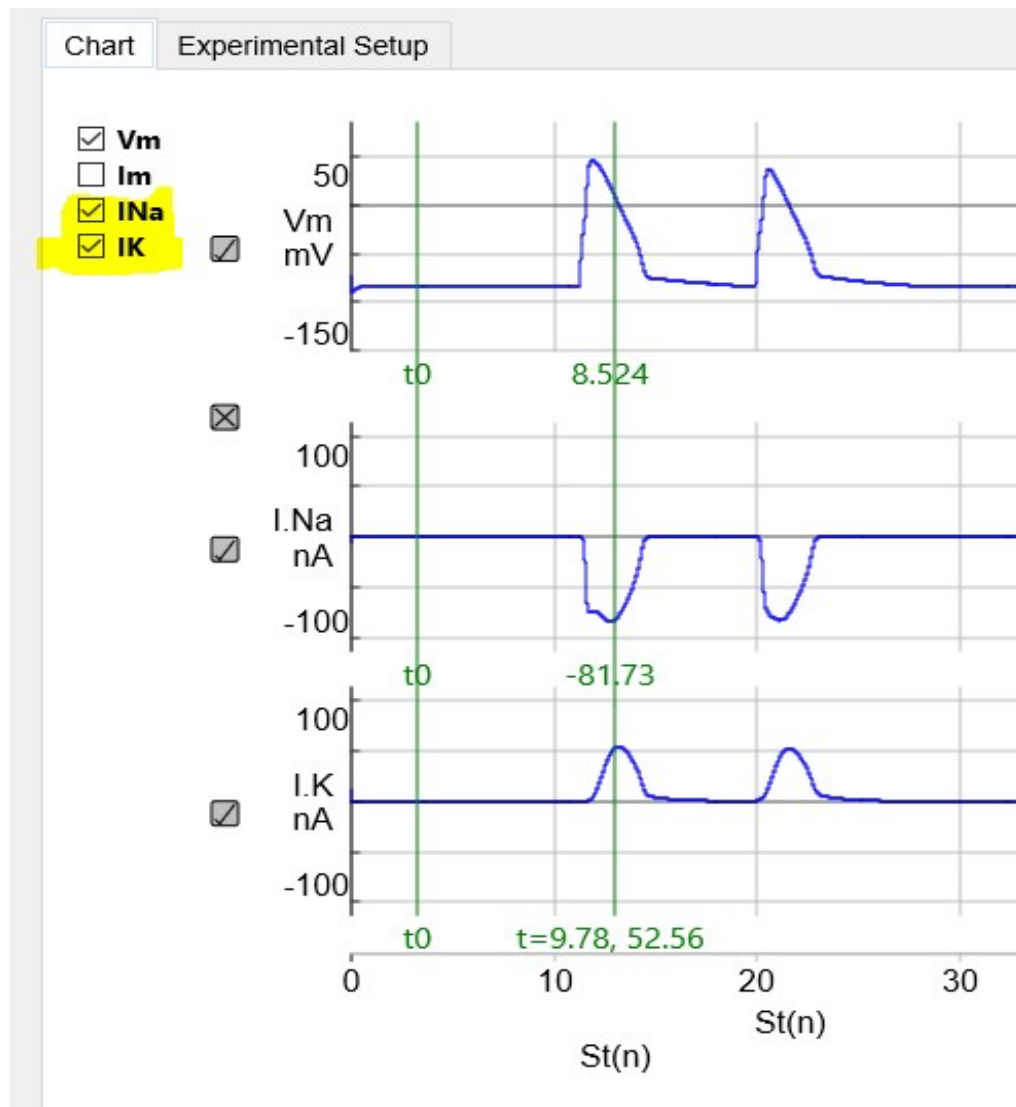
- Click the **Stop** button to stop recording.
- Using the scroll bar at the bottom of the chart recorder display, select a section of the recording containing the tissue contraction to be measured.
- You can expand or contract the display horizontally by change the **Window Size**.
- You can expand the display vertically by holding down the left-mouse button and indicating the vertical region with the mouse.



- Drag the green vertical measurement cursor on the chart display to the point on the recording trace to be measured. The membrane potential at the cursor point (in units of mV.) is displayed below the cursor. A second cursor (t_0) can be used to measure the time differences between two events.

Viewing Transmembrane Currents >

To view and measure the sodium and potassium trans-membrane currents which underlie the action potential, enable the **INa** and **IK** display options at the top-left of the display window.



The amplitude of the currents can be read using the green vertical readout cursor/

Printing >

To print a copy of the displayed recording on a printer, select **Print** from the **File** menu.

To select a specific printer or change printer settings, select **Printer Setup** from the **File** menu.

Copying to the Clipboard >

To copy the data points of the displayed recording to the Windows clipboard for pasting into a spreadsheet or graph plotting program, select **Copy Data** from the **Edit** menu.

To copy a picture of the displayed recording to the Windows clipboard for pasting into a Word document or a PowerPoint presentation, select **Copy Image** from the **Edit** menu.

Saving a recording to file

To save the recording for the current experiment to a data file, select **Save Experiment** from the **File** menu and enter the name of a new data file.

Loading an existing recording from file

To load a previously saved recording from a data file, select **Load Experiment** from the **File** menu and enter the name of the data file.

Acknowledgements

NMJSim was developed by the author while working within the Neuromuscular Research Group in the Department of Physiology & Pharmacology (subsequently the Strathclyde Institute for Pharmacy & Biomedical Sciences) at the University of Strathclyde in 2006 and updated 2026. The simulation model is based upon data from the experimental work of this group and from the studies and reviews listed in the references. Discussions and feedback from Ian G. Marshall, Bill Bowman, Alasdair Gibb and others are gratefully acknowledged.

NMJSim was created using the Embarcadero Delphi XE software development environment.

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